Siam Rahman Karip

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About

I am an independent researcher interested in large language models, spanning their evaluation, interpretability, and behavior across languages and settings. I remain open to a range of directions within this space and enjoy discovering new questions as I go.

Education

- **B.Sc in Computer Science and Engineering** *Feb 2021 - Jan 2025*
  BRAC University Dhaka, Bangladesh
  - CGPA: 3.59/4.00 (Distinction)

Work Experience

- **Research Assistant (RA)** *April 2025 – July 2026*
  United International University Dhaka, Bangladesh
  Supervisor: Mr. Nahid Hossain, *Assistant Professor and UG Program Coordinator*
  - Conducting research in large language models and natural language processing, with a focus on understanding model behavior, evaluation, and reliability.
  - Conducting occasional classes, quizzes, and exam-script grading.
- **Private Tutoring** *July 2022 – May 2025*
  - Tutored university-level students in Python programming, covering core concepts and practical problem-solving.
  - Assisted A-Level students in Pure Mathematics and Statistics, strengthening analytical and quantitative skills.

Research Interests

Large Language Models (LLMs), Natural Language Processing (NLP), Trustworthy Machine Learning

Publications

C=Conference, W=Workshop

[C.1] Nahid Hossain*, **Siam Rahman Karip***, Md Faisal Kabir, Swakkhar Shatabda and Salekul Islam. 2026. **Systematic Failure Analysis of Open-Weight LLM Annotators for Classification Tasks: Taxonomy, Uncertainty Signals, and Deployment Boundaries**. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)* (To Appear), Budapest, Hungary. Association for Computational Linguistics. (*Equal contribution)

[W.1] **Siam Rahman Karip** and Nahid Hossain. 2026. **AlienAnnotators at PsyDefDetect: What Lies Between the Lines: Probing Lightweight Open-Source LLMs for Psychological Defense Mechanism Detection**. In *Proceedings of the of the BioNLP 2026 (Shared Tasks)*, pages 213–223, San Diego, California, USA. Association for Computational Linguistics. DOI: 10.18653/v1/2026.bionlp-2.28

Research Projects

- **Evaluation Protocol and Decoding Paradigm Effects on Semantic Reasoning in Diffusion vs.Autoregressive Language Models** *In Progress* *Target: ACL*
  - Benchmarking diffusion and autoregressive language models across multiple evaluation protocols and decoding configurations on sentence-pair classification tasks requiring semantic reasoning.
  - Investigating how decoding paradigm and evaluation protocol choice jointly affect measured model performance and robustness, with a controlled comparison isolating architecture from training confounds.
- **Adapting Diffusion Language Models for Low-Resource Languages** *In Progress* *Target: ACL*
  - Investigating the applicability of diffusion-based language modeling approaches to low-resource language settings, where training data scarcity poses distinct challenges.
  - Exploring architectural and training adaptations to improve data efficiency and downstream task performance relative to standard autoregressive baselines.
- **EEG and Eye-Tracking Feature Contextualization based Affective Computing through LightGBM enabled Stacked Ensembling of LLMs** *Undergraduate Thesis* *BRAC University*
  - Introduced specific contextualized formatting for optimal tokenization.
  - Used LLMs (FLAN T5, GPT-2 and BERT) for classifying multimodal physiological numeric data.
  - Merged predictions from multiple LLMs using Majority Voting and Stacked Ensemble Learning Methods.

Honors and Awards

- **VC's List and Dean's List**
  BRAC University
  - Appeared on BRAC University's VC's list and Dean's list for last 3 semesters consecutively.
- **Academic Excellency Award**
  Uttara North City Corporation
  - Awarded for achieving a perfect GPA of 5.0 on all subjects in the Secondary School Board examination.

Courses and Certifications

- [🔗] **Transformer Models with Pytorch** *Datacamp*
- [🔗] **Large Language Models (LLMs) Concepts** *Datacamp*
- [🔗] **AI Fundamentals** *Datacamp*
- [🔗] **Intermediate Python** *Datacamp*

Skills

- **Programming Languages:** Python, C, Java, SQL, HTML, CSS, LaTeX
- **Tools for Deep Learning:** PyTorch (CUDA), TensorFlow, Transformers (Hugging Face), BitsandBytes
- **LLM Tooling and Evaluation:** Hugging Face Datasets, Hugging Face Evaluate, Scikit-learn
- **Data Science and Analytics:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, OpenCV
- **Version Control and Environment:** Git, GitHub

Projects

- [ ] **Cross-Task Capability Testing of SLMs**
- [ ] **Sports Image Classification using a custom CNN model**
- [ ] **Email Spam Classification using ML models**
- [ ] **Credit Card Fraud Detection using ML models**

References

1. **Nahid Hossain**
   Assistant Professor and Program Coordinator
   Department of Computer Science and Engineering
   United International University
   Email: nahid@cse.uiu.ac.bd
   Relationship: Employer, Project Supervisor

2. **Swakkhar Shatabda**
   Professor
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   BRAC University
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   Relationship: Project Co-supervisor

3. **Rafeed Rahman**
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   Relationship: Thesis Supervisor